

MBFD Mid-Mount Ladder Workgroup — Final Equipment Selection & Implementation Report

Miami Beach Fire Department — Q1 2026

Mid-Mount Ladder Workgroup

2026-03-18

Executive Summary

This report documents the **final outcome** of the Miami Beach Fire Department Mid-Mount Ladder Workgroup evaluation process (Q1 2026). Following structured multi-day evaluations of hydraulic rescue tools, cut-off saws, vehicle stabilization systems, and specialty tools, the workgroup has reached consensus on final equipment selections.

This report is submitted to **Fire Chief Digna Abello** and the **Health & Safety Committee**.

Final Selections Summary

Category	Selected Equipment	Platform
Cut-Off Saw	DeWalt 12" Battery Cut-Off Saw	Frontline
Chainsaw	DeWalt 18" Chain-saw	Frontline
Stabilization	4 × Holmatro V-Struts	Frontline
Spreader	Hurst SP 777 E3 Connect	CAPTUM
Cutter	Hurst S 789 E3 Connect	CAPTUM
Ram	Hurst CR 522 E3 Connect	CAPTUM
Specialty (Trial)	Holmatro T1	Trial
Heavy Extrication	Hurst M40 (40" spreader)	300 (BC)
Future Addition	Lifting Struts (pending)	300 / Capt 5

Evaluation Framework

The workgroup employed a standardized scoring model across all 14 products from 6 manufacturers.

Scoring Dimensions:

Dimension	Description
Capability	Force output, cutting capacity, spreading distance
Usability	Ergonomics, control interface, operator comfort
Maintainability	Service requirements, durability, field-repairability
Deployability	Storage footprint, activation speed, weight

Each dimension scored 0–100. Composite **Overall Score** computed as weighted aggregate.

Brand Performance

Brand	Products	Avg. Score
DeWalt	1	91.25
Holmatro	6	87.93
Hurst	4	84.60
Paratech	1	76.13
Makita	1	64.83
Husqvarna	1	61.78

Holmatro vs. Hurst Dimension Comparison

Dimension	Holmatro	Hurst	Δ
Capability	89.41	87.56	+1.85
Usability	93.99	85.76	+8.23
Maintainability	89.82	82.58	+7.24
Deployability	89.69	82.74	+6.95

Holmatro leads all four dimensions, largest gap in **Usability** (+8.23 pts).

Frontline Tools

Spreaders

Product	Score	Cap.	Usab.	Maint.	Deploy.
Holmatro PSP40	92.02	89.41	93.99	89.82	89.69
Hurst SP 777 E3	90.99	87.56	85.76	82.58	82.74

Cutters

Product	Score	Cap.	Usab.	Maint.	Deploy.
Holmatro PCU30CL	88.86	89.41	93.99	89.82	89.69
Hurst S 789 E3	82.57	87.56	85.76	82.58	82.74

Rams

Product	Score	Cap.	Usab.	Maint.	Deploy.
Holmatro PRA40	91.29	89.41	93.99	89.82	89.69
Hurst CR 522 E3	86.02	87.56	85.76	82.58	82.74

Cut-Off Saws

Product	Score	Blade	RPM
DeWalt DCPS612AG2	91.25	12"	—
Makita GEC01PL4	64.83	14"	5,300
Husqvarna K1 Pace	61.78	14"	3,400

DeWalt outperformed 14" competitors by 26–29 points despite smaller blade.

Vehicle Stabilization

Product	Score	Mechanism	Weight
Holmatro V-Strut	87.28	Auto-lock	15.87 lbs
Holmatro OmniShore	85.87	Pneumatic	—
Paratech StrutDriver	76.13	Manual	22.57 lbs

V-Strut auto-lock enables 15-second setup at 15.87 lbs.

Specialty Tools

Holmatro T1 — Score: 82.23

Six-function tool (cut, wedge, ram, spread, hammer, lift) at 17 lbs. Trial basis as Rabbit tool replacement candidate.

Hurst M40 — Score: 78.80

40" heavy spreader for 300 (Battalion Chief) vehicle. Supplemental heavy rescue capability.

Apparatus Deployment

Frontline (L1 / L3)

All Hurst E3 CAPTIUM tools + 4× V-Struts + DeWalt saw/chainsaw + T1 trial.

Command Level (300 / Captain 5)

Hurst M40 + batteries + charging station + future lifting struts.

Training Plan

Train-the-Trainer Model:

1. Vendor training for workgroup SMEs
2. Shift-based department rollout (no overtime)
3. Deep familiarization on all systems

All training completed before apparatus enters service.

Data Source Verification

All values sourced from Product_Data_Master.csv (14 products, 31 columns). No interpolation or external sources.

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